# Dairy Mate: Cattle Mastitis Detection

Round 1 Submission Report


### Team Members

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## 1. Overview

This project implements an end-to-end machine learning pipeline for detecting mastitis in cattle from udder photographs. The pipeline covers dataset annotation, udder segmentation, inflammation-region highlighting, preprocessing, and classification of each image as Healthy or Mastitis.

## 2. Dataset and Annotation

The dataset consists of 100 images: 25 healthy udder images and 75 mastitis-affected udder images, sourced into Dataset/healthy_images/ and Dataset/Mastitis_images/. Manually drawing segmentation masks for 100 images was not practical within the timeframe, so the udder region in each image was annotated using a zero-shot auto-labeling pipeline (src/auto_annotate.py):

- Grounding DINO (IDEA-Research/grounding-dino-tiny) locates the udder in each image using the text prompt "udder .", producing a bounding box (a fixed fallback region is used if detection confidence is below threshold).
- Segment Anything Model (facebook/sam-vit-base) takes that bounding box and produces a precise pixel-level mask of the udder.
- The mask outline is simplified into a polygon and saved in YOLO-segmentation label format, split 80/20 into train and validation sets.

This produced consistent udder labels across the dataset without manual annotation, while still requiring no more than a bounding-box-level text prompt per image.

## 3. Segmentation and Inflammation Mapping

A YOLOv8n-seg model was trained on the auto-generated udder masks (src/train_yolo.py, 30 epochs, image size 640, batch size 16) and saved as Model/best_udder_yolo.pt. At inference time, this model isolates the udder region in a new image and excludes background clutter such as legs and flooring.

To highlight the mastitis-affected area itself, an HSV color-thresholding step (segmentation.py) is applied within the segmented udder region only. Two red-hue ranges are combined to account for the wraparound of the hue channel, and morphological opening removes noise. This produces a red hotspot mask over inflamed skin, and an Inflammation Index is computed as the percentage of udder pixels that are classified as inflamed. This index and the corresponding overlay image are the outputs used to visually and numerically flag the mastitis-affected area for each image.

## 4. Preprocessing Pipeline

Each image proceeds through the following steps before reaching the classifier:

- Segmentation: the trained YOLOv8n-seg model produces the udder mask and bounding box.
- Cropping: the image is cropped to the udder bounding box, removing irrelevant background.
- Metadata compilation: src/annotate.py records the bounding box and inflammation index for every image into Dataset/annotations.json.
- Resizing and normalization: crops are resized to 224x224 and normalized to ImageNet mean and standard deviation.
- Augmentation (training only): random horizontal flip, random rotation up to 15 degrees, and color jitter on brightness, contrast, and saturation, used to reduce overfitting on a small dataset.

# 5. Model Selection and Justification

A two-stage architecture was used: YOLOv8n-seg for segmentation, followed by two parallel classifiers, ResNet-50 and MobileNet-V3-Large, both fine-tuned from ImageNet-pretrained weights (src/train.py) rather than trained from scratch.

## 5.1 Why this architecture

- Two-stage over end-to-end: segmenting the udder before classification removes background and irrelevant anatomy, so the classifier only has to reason about the region that matters. This is especially useful with a small dataset.
- Transfer learning over training from scratch: with only 100 images, training a deep CNN from scratch would overfit heavily. Fine-tuning ImageNet-pretrained ResNet-50 and MobileNet-V3 gives usable accuracy with far fewer labeled examples.
- ResNet-50: chosen as a stronger accuracy benchmark, given its depth and well-established performance on image classification tasks.
- MobileNet-V3-Large: chosen as a lightweight, low-latency alternative suited to on-farm or edge deployment, where compute may be limited. Offering both models lets a user trade off accuracy against speed.
- YOLOv8n-seg: the nano variant was chosen for fast inference and because Ultralytics' segmentation head natively supports the polygon-style masks produced by the auto-labeling step.

## 5.2 Challenges Faced

- Low dataset size and class imbalance: only 100 images total, and unevenly split (25 healthy, 75 mastitis). This risked overfitting and bias toward the majority class. This was addressed through transfer learning instead of training from scratch, data augmentation applied only to the training split, use of comparatively lightweight architectures, and saving only the checkpoint with the best validation accuracy rather than the final epoch.
- Annotation problem, specifically how to mark the mastitis-affected area: there was no manually labeled ground truth for diseased tissue, and hand-drawing segmentation masks for 100 images was not feasible in the available time. This was addressed in two parts: automating udder localization itself using zero-shot Grounding DINO and SAM instead of manual polygon drawing, and approximating the mastitis-affected area using HSV-based redness detection constrained to the segmented udder, producing both a visual hotspot overlay and a quantitative inflammation index. This inflammation index is a practical proxy rather than a clinically verified ground truth, which is a known limitation of the current approach.

## 5.3 Training Results

Both classifiers were trained for up to 30 epochs, with the checkpoint saved whenever validation accuracy improved. MobileNet-V3 reached its best validation accuracy of 90.0% by epoch 19 (train accuracy 100.0%, loss 0.0229) and held that accuracy through epoch 22 before validation dipped in later epochs. ResNet-50 reached its best validation accuracy of 90.0% at epoch 4 (train accuracy 95.0%, loss 0.1388), matching that peak again at epoch 25; its training accuracy continued climbing to 100.0% in between, while validation accuracy settled around 80.0% for much of the run, indicating mild overfitting between these peaks. In both cases, saving only the best-validation checkpoint, rather than the final epoch, was the mitigation used to avoid deploying an overfit model.

## 5.4 Hardware

Both classifiers and the YOLOv8n-seg model were trained locally on an NVIDIA RTX 3050 (6 GB VRAM) laptop GPU. A larger data-center GPU such as an NVIDIA Tesla T4 (16-30 GB VRAM) was considered but not used, since the dataset is only 100 images and both classifiers are lightweight (ResNet-50 and MobileNet-V3-Large) rather than large-scale models; the RTX 3050 trained each model in a reasonable time, and a T4-class GPU would be unnecessary compute for this task's scale.

# 6. Conclusion

The resulting pipeline takes a raw udder photograph through automated annotation, learned segmentation, inflammation highlighting, and classification, and is served through an interactive Gradio dashboard that lets a user choose between the two trained classifiers and view the intermediate computer vision outputs alongside the final Healthy or Mastitis diagnosis.

The live demo is deployed on Hugging Face Spaces using the free CPU Basic tier (2 vCPU, 16 GB RAM, no GPU), so all inference in the hosted demo runs on CPU only. This makes the lightweight MobileNet-V3 option particularly relevant for the deployed version, and is also why hf_space/requirements.txt pins CPU-only PyTorch and opencv-python-headless builds.